

CBE 3610 Handbook

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All information in this Reference Book is adapted from or inspired by the following source except where indicated otherwise:

Fogler, H. S., *Elements of Chemical Reaction Engineering*, 5th Edition, Pearson Education, Inc., Boston, 2016, ISBN: 978-0-13-388751-8.

1 Purpose of This Handbook:

This course is built around one central idea: you learn chemical reaction engineering by *engaging with it*, not by memorizing equations. The goal of the CBE 3610 Handbook is therefore not to replace understanding. Rather, its purpose is to help you focus on the parts of reactor design that matter most: identifying the governing principles, choosing an appropriate approach, making reasonable assumptions, carrying out the analysis, and interpreting the results.

1. I do not expect you to memorize every equation used in chemical reaction engineering. Research consistently shows that learning is deeper when students focus on concepts, problem-solving, and engineering reasoning rather than rote memorization. Consequently, this Handbook contains the core equations, definitions, variable descriptions, and relationships that we use throughout the course. Rather than memorizing formulas, I encourage you to learn where equations come from, when they apply, what assumptions they require, and how they connect to the governing balances.
2. In chemical reaction engineering, knowing *how to start* is often more important than remembering a specific equation. When students become stuck, the solution is usually to identify the reactor type, determine whether equilibrium matters, determine whether temperature effects matter, and return to the governing mole and energy balances. This Handbook is designed to help guide that process by collecting the most commonly used equations and concepts in one place.
3. I also believe strongly in affordable and accessible learning. The course textbook is available to Ohio State students at no cost through the university libraries, and purchasing a textbook is

not required. As a result, I wanted to provide a resource that contains the fundamental information needed to succeed in the course regardless of whether you have the textbook open in front of you. In addition to equations, this Handbook includes definitions, explanations, units, and clarifications that I hope make the material more approachable.

4. Finally, this Handbook is intended to support the development of engineering judgment. Real reactor design problems rarely begin with an obvious path forward. The challenge is not simply solving equations; it is choosing an appropriate model, understanding its assumptions and limitations, and deciding whether the result is physically reasonable. I hope this Handbook helps you build that confidence throughout the semester.

When you are unsure where to start, open this book, identify the governing balances, and take the next reasonable step. You do not need to understand everything immediately. If you engage consistently, ask questions, and keep moving forward, you can succeed in chemical reaction engineering.

Happy reactor designing!

— RBG

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2 Common CRE Problem Types

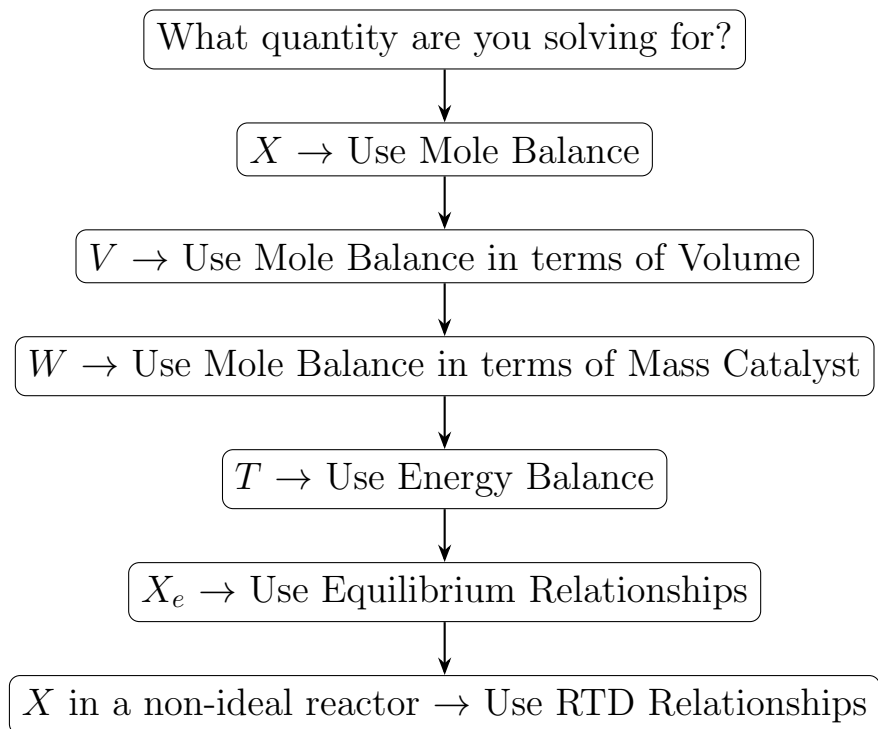


Figure 1: Common problem types in CBE 3610.

3 How Do I Start?

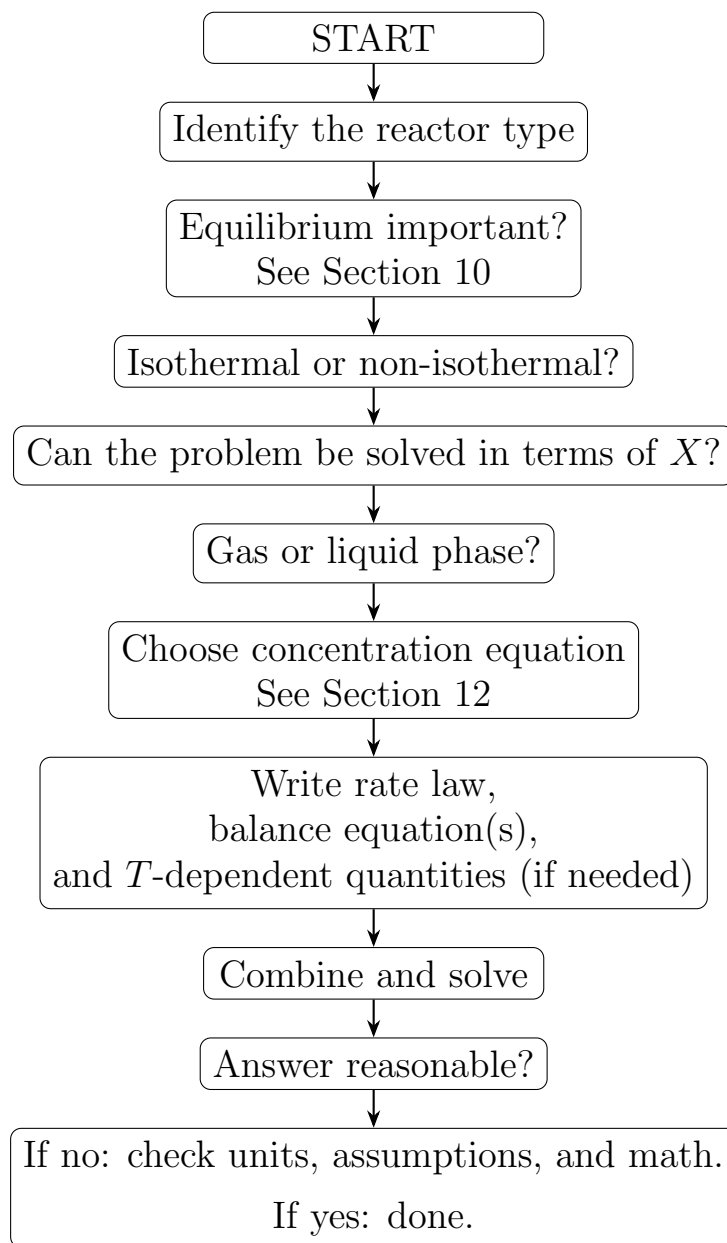


Figure 2: General problem solving workflow for CBE 3610.

4 How Do I Solve CRE Problems?

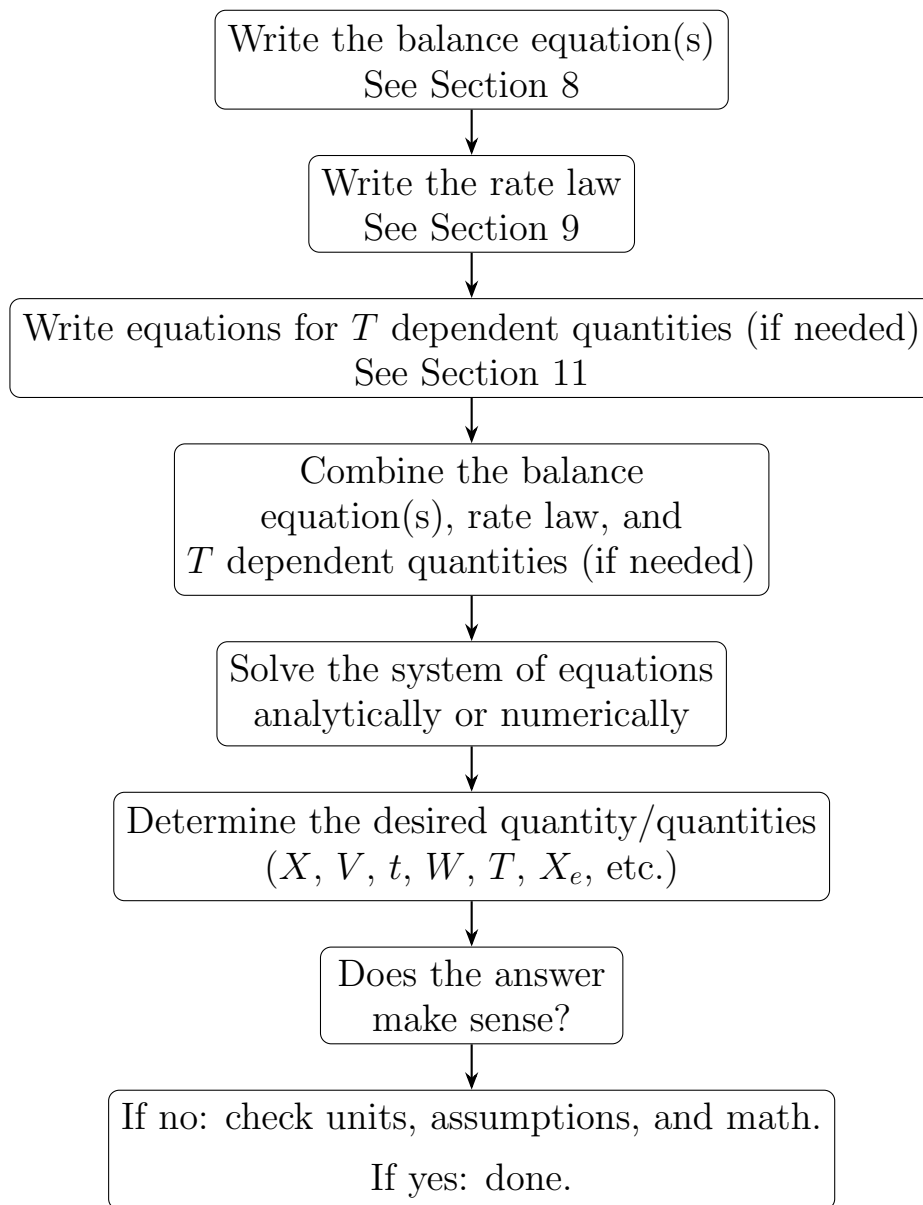


Figure 3: General solution procedure for reactor-design problems.

5 Definitions/Descriptions of Useful CRE Quantities and Terms

5.1 Gas Constant Values

$$R = \frac{0.73 \text{ ft}^3 \cdot \text{atm}}{\text{lbmol} \cdot ^\circ\text{R}} \quad R = \frac{1.987 \text{ Btu}}{\text{lbmol} \cdot ^\circ\text{R}}$$

$$R = \frac{8.314 \text{ dm}^3 \cdot \text{kPa}}{\text{mol} \cdot \text{K}} \quad R = \frac{8.314 \text{ J}}{\text{mol} \cdot \text{K}}$$

$$R = \frac{0.082 \text{ dm}^3 \cdot \text{atm}}{\text{mol} \cdot \text{K}} \quad R = \frac{1.987 \text{ cal}}{\text{mol} \cdot \text{K}}$$

Boltzmann's constant: $k_B = 1.381 \times 10^{-23} \frac{\text{J}}{\text{molecule} \cdot \text{K}}$

5.2 Other Useful Definitions and Descriptions

The limiting reactant is often denoted "LR".

The conversion of B, $X_B = \frac{\text{Moles of B reacted}}{\text{Moles of B fed or present initially}}$.

Residence time, $\tau = \frac{\text{system volume}}{\text{inlet volumetric flowrate}} = \frac{V}{v_0}$

The change in the total number of moles per mole of limiting reactant (LR) reacted, $\delta = \frac{\text{Change in the total number of moles}}{\text{Mole of LR reacted}}$.

Ratio of molar amounts or molar flowrates, $\Theta_B = \frac{\text{Moles of species B fed or present initially}}{\text{Moles of LR fed or present initially}}$.

Parameter for reactions for which there is a change in the total number of moles: $\epsilon = \frac{\text{Change in total number of moles for complete conversion}}{\text{Total moles fed}}$.

The Damköhler number, $\text{Da} = \frac{\text{Rate of reaction}}{\text{Rate of convection}}$.

The instantaneous selectivity, $S_{D/U} = \frac{r_D}{r_U}$.

The overall selectivity, $\tilde{S}_{D/U} = \frac{F_D|_{\text{exit}}}{F_U|_{\text{exit}}}$.

The instantaneous yield, $Y_D = \frac{r_D}{-r_{LR}}$.

The overall yield, $\tilde{Y}_D = \frac{\text{Moles of desired product formed}}{\text{Moles of LR reacted}}$.

The definition of *parallel* is: Things that happen side by side (and, for the most part, independently). The definition of *serial* is: Things that happen in series. When events happen in series, each individual event depends in at least some way on the events prior.

An *isothermal process* maintains constant temperature. An *isobaric process* maintains constant pressure. An *isochoric process* maintains constant volume. An *adiabatic process* is one in which there is no heat transfer between the system and its surroundings.

The *heat generated* term, Q_g or $G(T)$, is due to the reaction. E.g., $G(T)$ for a CSTR carrying out a reaction in the form $A + \text{other reactants} \longrightarrow \text{products}$ can be written as equal to $(-\Delta H_{rxn}^\circ) \frac{-r_A V}{F_{A0}}$.

The *heat removed* term, Q_r or $R(T)$, is due to 1) the heat exchanger, and 2) the sensible and/or latent heat changes that bring the feed to the reaction temperature.

The *ignition temperature* for a tank reactor is the feed temperature at which a fraction of a degree increase would cause a jump in the reactor temperature from the lower steady state temperature to the upper steady state temperature. The *extinction temperature* is the feed temperature at which a fraction of a degree decrease would cause a jump in the reactor temperature from the upper steady state temperature to the lower steady state temperature.

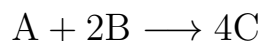
Quantities with a dot over them are rates. E.g., $\dot{Q}$ is heat flow per unit time, and has units of, e.g., J/s.

Quantities with a “0” subscript are for inlet or initial conditions. E.g., T_0 for a CSTR is the temperature of the feed. T_0 for a batch reactor is the temperature initially.

6 Heat of Reaction and Heat Capacity

6.1 Units

The following formulas require the heat of reaction, ΔH_{rxn}° . The units of ΔH_{rxn}° are typically expressed as energy per mole, where the “mole” refers to a specific chemical species. For example, consider the reaction



If

$$\Delta H_{rxn}^\circ = -100 \text{ kJ/mol A reacted,}$$

then the same heat of reaction can also be expressed as

$$-50 \text{ kJ/mol B reacted}$$

or

$$-25 \text{ kJ/mol C formed.}$$

When using the energy balance, the mole units associated with ΔH_{rxn}° must be consistent with those used in the reaction-rate term because these quantities are multiplied together. For example, if the mole balance and rate law are written in terms of species A, then the rate will

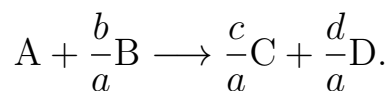
typically be expressed as r_A , which has units of moles of A per volume per time. In this case, ΔH_{rxn}° should be expressed in units of energy per mole of A:

$$(-r_A)(-\Delta H_{rxn,A}) = \left(\frac{\text{mol A}}{\text{volume} \cdot \text{time}} \right) \left(\frac{\text{kJ}}{\text{mol A}} \right) = \frac{\text{kJ}}{\text{volume} \cdot \text{time}}$$

A convenient way to obtain ΔH_{rxn}° in the proper units is to rewrite the reaction so that the stoichiometric coefficient of the species appearing in the rate expression is equal to one. For example, if the rate is written in terms of species A, then rewrite



as



The resulting value of ΔH_{rxn}° will then have units of energy per mole of A reacted.

$$\Delta H_{rxn,A}^\circ = \left(\frac{d}{a}H_D^\circ + \frac{c}{a}H_C^\circ - \frac{b}{a}H_B^\circ - H_A^\circ \right) \quad (1)$$

where the H° are heats of formation measured at the reference temperature T_{ref} . Calculated this way, $\Delta H_{rxn,A}^\circ$ is in terms of moles of A and thus has units of energy per mole A, e.g., kJ/mol A.

Similarly,

$$\Delta C_{P,A} = \left(\frac{d}{a}C_{P,D} + \frac{c}{a}C_{P,C} - \frac{b}{a}C_{P,B} - C_{P,A} \right) \quad (2)$$

gives the change in heat capacity per mole A reacted.

6.2 As Functions of T

H are related to C_P by

$$H(T) - H(T_{ref}) = \int_{T_{ref}}^T C_P(T) dT \quad (3)$$

If we can assume that C_P is constant (i.e., that it does not depend on T), then

$$H(T) = H(T_{ref}) + C_P(T - T_{ref}) \quad (4)$$

Combining Equations 1, 2, and 4

$$\Delta H_{rxn}(T) = \Delta H_{rxn}^\circ(T_{ref}) + \Delta C_P(T - T_{ref}) \quad (5)$$

7 Levenspiel Plot Equations

For a CSTR:

$$V = \left(\frac{F_{LR,0}}{-r_{LR}(X)} \right) X \quad (6)$$

In other words, to estimate CSTR volume using a Levenspiel plot, multiply the y value (i.e., $\left(\frac{F_{LR,0}}{-r_{LR}(X)} \right)$) by the x value (i.e., X).

For a PFR:

$$V = F_{LR,0} \int_0^X \frac{dX}{-r_{LR}(X)} \quad (7)$$

In other words, to estimate PFR volume using a Levenspiel plot, integrate $\left(\frac{F_{LR,0}}{-r_{LR}(X)} \right)$ with respect to X .

In both cases, X must be taken with respect to the limiting reactant.

8 Common Reactors

The energy balances in this section assume a heat exchanger with constant T_a . For other heat exchange scenarios, see Section 13.

8.1 Continuously stirred tank reactor (CSTR)

8.1.1 Mole Balance

In terms of F :

$$V = \frac{F_{A0} - F_A}{-r_A} \quad (8)$$

In terms of X :

$$V = \frac{F_{LR,0} (X_{\text{out}} - X_{\text{in}})}{(-r_{LR})_{\text{out}}} \quad (9)$$

When written this way, X must be written with respect to the limiting reactant.

This equation can be used to solve for X . The resulting equation is sometimes denoted X_{MB} .

8.1.2 Energy Balance

Note that the equations below neglect the $\dot{W}_s$ term. In some circumstances (e.g., viscous liquid fluid), this term may be necessary.

In terms of F :

$$V = \frac{UA(T_a - T) + \sum_j F_{j0} C_{Pj} (T_0 - T)}{(-r_B) (-\Delta H_{rxn,B}(T))} \quad (10)$$

Note: See Section 6.1. The $(-r_B)$, $(-\Delta H_{rxn,B})$, and, where applicable, $\Delta C_{P,B}$ terms can be expressed on the basis of any species, provided all three quantities are expressed on the same stoichiometric basis.

Multiple reactions (q reactions and m species):

$$V = \frac{UA(T_a - T) + \sum_j^m F_{j0} C_{Pj} (T_0 - T)}{\sum_i^q (-r_{iB}) (-\Delta H_{rxn,iB}(T))} \quad (11)$$

Note: See Section 6.1. For each reaction i , r_{iB} , $\Delta H_{rxn,iB}$, and, where applicable, $\Delta C_{P,iB}$ can be expressed on the basis of any species participating in reaction i , provided all three quantities for that reaction are expressed on the same stoichiometric basis.

In terms of X :

$$X_{EB} = \frac{\left(\frac{UA}{F_{LR,0}} (T_a - T) \right) + \sum_i \Theta_i C_{Pi} (T_0 - T)}{\Delta H_{rxn,LR}(T)} \quad (12)$$

where $\Delta H_{rxn,LR}$ and, where applicable, $\Delta C_{P,LR}$ should be written with respect to the limiting reactant.

Note that this equation neglects the $\dot{W}_s$ term. In some circumstances (e.g., viscous liquid fluid), this term may be necessary.

The heat generated, $G(T)$ and heat removed, $R(T)$ are obtained by multiplying X_{MB} and X_{EB} , respectively, by $(-\Delta H_{rxn})$.

8.2 Plug flow reactor (PFR)

8.2.1 Mole Balance

In terms of F :

$$\frac{dF_A}{dV} = r_A \quad (13)$$

In terms of X :

$$F_{LR,0} \frac{dX_{LR}}{dV} = -r_{LR} \quad (14)$$

When written this way, X must be written with respect to the limiting reactant.

8.2.2 Energy Balance

In terms of F :

$$\frac{dT}{dV} = \frac{(-r_B)(-\Delta H_{rxn,B}(T)) + Ua(T_a - T)}{\sum_i F_i C_{Pi}} \quad (15)$$

Note: See Section 6.1. The $(-r_B)$, $(-\Delta H_{rxn,B})$, and, where applicable, $\Delta C_{P,B}$ terms can be expressed on the basis of any species, provided all three quantities are expressed on the same stoichiometric basis.

Multiple reactions (q reactions and m species):

$$\frac{dT}{dV} = \frac{\sum_i^q (-r_{iB})(-\Delta H_{rxn,iB}(T)) + Ua(T_a - T)}{\sum_j^m F_j C_{Pj}} \quad (16)$$

Note: See Section 6.1. For each reaction i , $(-r_{iB})$, $(-\Delta H_{rxn,iB})$, and, where applicable, $\Delta C_{P,iB}$ can be expressed on the basis of any species participating in reaction i , provided all three quantities for that reaction are expressed on the same stoichiometric basis.

In terms of X :

$$\frac{dT}{dV} = \frac{(-r_{\text{LR}}) (-\Delta H_{\text{rxn}}(T)) + Ua(T_a - T)}{F_{\text{LR},0} \left(\sum_j \Theta_j C_{\text{P}j} + \Delta C_{\text{P}} X \right)} \quad (17)$$

where $\Delta H_{\text{rxn,LR}}$ and, where applicable, $\Delta C_{\text{P,LR}}$ should be written with respect to the limiting reactant.

8.3 Batch Reactor

8.3.1 Mole Balance

in terms of N :

$$\frac{dN_{\text{A}}}{dt} = r_{\text{A}} V \quad (18)$$

in terms of X :

$$N_{\text{LR},0} \frac{dX_{\text{LR}}}{dt} = -r_{\text{LR}} V \quad (19)$$

When written this way, X must be written with respect to the limiting reactant.

8.3.2 Energy Balance

in terms of N , single reaction:

$$\frac{dT}{dt} = \frac{(-r_{\text{B}} V) (-\Delta H_{\text{rxn,B}}(T)) + U A (T_a - T)}{\sum_j N_j C_{\text{P}j}} \quad (20)$$

Note: See Section 6.1. The $(-r_{\text{B}})$, $(-\Delta H_{\text{rxn,B}})$, and, where applicable, $\Delta C_{\text{P,B}}$ terms can be expressed on the basis of any species, provided all three quantities are expressed on the same stoichiometric basis.

Multiple reactions (q reactions and m species):

$$\frac{dT}{dt} = \frac{V \sum_{i=1}^q (-r_{iB}) (-\Delta H_{rxn,iB}(T)) + UA(T_a - T)}{\sum_{j=1}^m N_j C_{Pj}} \quad (21)$$

Note: See Section 6.1. For each reaction i , $(-r_{iB})$, $(-\Delta H_{rxn,iB})$, and, where applicable, $\Delta C_{P,iB}$ can be expressed on the basis of any species participating in reaction i , provided all three quantities for that reaction are expressed on the same stoichiometric basis.

in terms of X :

$$\frac{dT}{dt} = \frac{V(-r_{LR})(-\Delta H_{rxn}(T)) + UA(T_a - T)}{N_{LR,0} \left(\sum_j \Theta_j C_{Pj} + \Delta C_P X \right)} \quad (22)$$

where $\Delta H_{rxn,LR}$ and, where applicable, $\Delta C_{P,LR}$ should be written with respect to the limiting reactant.

8.4 Packed Bed Reactor (PBR)

8.4.1 Mole Balance

in terms of F :

$$\frac{dF_A}{dW} = r'_A \quad (23)$$

in terms of X :

$$F_{LR0} \frac{dX_{LR}}{dW} = -r'_{LR} \quad (24)$$

When written this way, X must be written with respect to the limiting reactant.

8.4.2 Energy Balance

in terms of F :

$$\frac{dT}{dW} = \frac{(-r'_B)(-\Delta H_{rxn,B}(T)) + \frac{U_a}{\rho_b}(T_a - T)}{\sum_j F_j C_{Pj}} \quad (25)$$

Note: See Section 6.1. The $(-r_B)$, $(-\Delta H_{rxn,B})$, and, where applicable, $\Delta C_{P,B}$ terms can be expressed on the basis of any species, provided all three quantities are expressed on the same stoichiometric basis.

Multiple reactions (q reactions and m species):

$$\frac{dT}{dW} = \frac{\sum_i^q (-r'_{iB})(-\Delta H_{rxn,iB}(T)) + \frac{U_a}{\rho_b}(T_a - T)}{\sum_j^m F_j C_{Pj}} \quad (26)$$

Note: See Section 6.1. For each reaction i , $(-r_{iB})$, $(-\Delta H_{rxn,iB})$, and, where applicable, $\Delta C_{P,iB}$ can be expressed on the basis of any species participating in reaction i , provided all three quantities for that reaction are expressed on the same stoichiometric basis.

in terms of X :

$$\frac{dT}{dW} = \frac{(-r'_{LR})(-\Delta H_{rxn}(T)) + \frac{U_a}{\rho_b}(T_a - T)}{F_{LR,0} \left(\sum_j \Theta_j C_{Pj} + \Delta C_{PX} \right)} \quad (27)$$

where $\Delta H_{rxn,LR}$ and, where applicable, $\Delta C_{P,LR}$ should be written with respect to the limiting reactant.

9 General Form of the Rate Law with Examples

Rate of consumption of species A:

$$-r_A = [k_A(T)] [\text{fn}(C_A, C_B, \dots)] \quad (28)$$

E.g., fn could follow a power law model, e.g.,

$$\text{fn}(C_A, C_B, \dots) = C_A^\alpha C_B^\beta \dots \quad (29)$$

For a single reactant:

$$-r_A = k C_A^m \quad (30)$$

For two reactants:

$$-r_A = k C_A^m C_B^m \quad (31)$$

For a reversible reaction:

$$-r_A = k_f \left(\prod_{\text{reactants}} C_i^{|\nu_i|} - \frac{\prod_{\text{products}} C_j^{|\nu_j|}}{K_C} \right) \quad (32)$$

where k_f is the forward rate constant, defined in terms of one of the reactant species, K_C is the equilibrium constant with respect to concentration, and ν are stoichiometric coefficients. We take the absolute values of ν (which are negative for reactants and positive for products). k_f and K_C must be defined in terms of the same species.

Example reversible reaction, $aA + bB \rightleftharpoons cC + dD$:

$$-r_A = k_{f,A} \left(C_A^a C_B^b - \frac{C_C^c C_D^d}{K_C} \right) \quad (33)$$

where K_C must be defined per mole of species A, since k_f is in terms of species A.

Relative rates: For the reaction $aA + bB \longrightarrow cC + dD$,

$$\frac{-r_A}{a} = \frac{-r_B}{b} = \frac{r_C}{c} = \frac{r_D}{d} \quad (34)$$

10 Equilibrium Relationships

For the reaction $A + \frac{b}{a}B \rightleftharpoons \frac{c}{a}C + \frac{d}{a}D$, the equilibrium constant in terms of pressure,

$$K_p = \frac{P_{C,eq}^{c/a} P_{D,eq}^{d/a}}{P_{A,eq} P_{B,eq}^{b/a}} \quad (35)$$

The equilibrium constant in terms of concentration:

$$K_C = \frac{C_{C,eq}^{c/a} C_{D,eq}^{d/a}}{C_{A,eq} C_{B,eq}^{b/a}} \quad (36)$$

In Equations 35 and 36, the subscript “eq” denotes that the partial pressures or concentrations must be at their equilibrium values.

For an ideal gas:

$$K_{eq} = \frac{K_p}{(1 \text{ bar})^\delta} \quad (37)$$

For an ideal liquid:

$$K_{eq} = \frac{K_C}{(1 \text{ M})^\delta} \quad (38)$$

K_{eq} is related to the Gibbs Free Energy of reaction via:

$$K_{eq} = \exp\left(\frac{-\Delta G_{rxn}^\circ}{RT}\right) \quad (39)$$

These expressions can be used to solve for the partial pressures or concentrations at equilibrium.

From thermodynamics,

$$\Delta G = \Delta H - T\Delta S \quad (40)$$

For an ideal gas:

$$K_p = K_c (RT)^\delta \quad (41)$$

Since liquids are incompressible and do not have partial pressures, K_p is meaningless for a liquid.

The approach to equilibrium, β

$$\beta = \frac{Q_C}{K_C} = \frac{Q_p}{K_p} \quad (42)$$

where Q_C and Q_p are the reaction quotients:

$$Q_C = \frac{C_C^{c/a} C_D^{d/a}}{C_A C_B^{b/a}} \quad (43)$$

and

$$Q_p = \frac{P_C^{c/a} P_D^{d/a}}{P_A P_B^{b/a}} \quad (44)$$

i.e., Q_C and Q_p involve the actual concentrations or partial pressures, while K_C and K_p involve the equilibrium values.

Determining if equilibrium will influence the reaction:

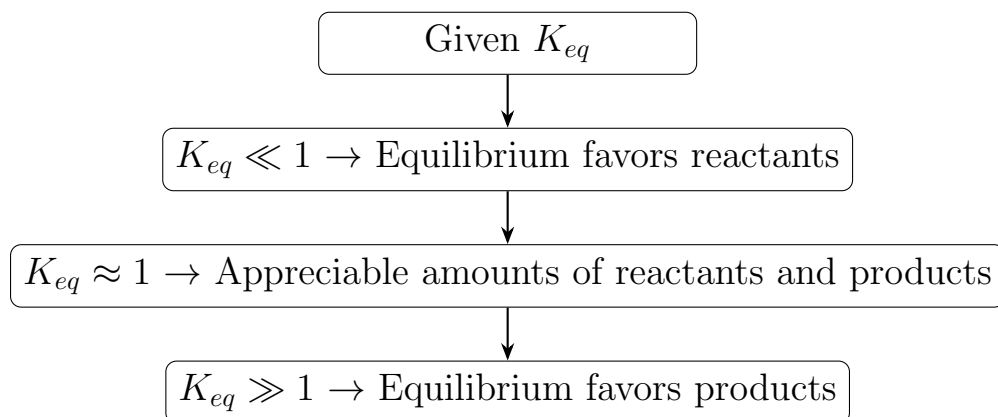


Figure 4: Using the magnitude of K_{eq} to assess the importance of equilibrium.

11 Temperature dependent quantities

Specific rate constant for species A (Arrhenius Equation):

$$k_A(T) = A \exp\left(-\frac{\Delta E_{act}}{RT}\right) \quad (45)$$

Temperature-dependence of rate constant (Arrhenius Equation):

$$k(T) = k(T_0) \exp\left[\frac{\Delta E_{act}}{R} \left(\frac{1}{T_0} - \frac{1}{T}\right)\right] \quad (46)$$

Note that this equation assumes that A is constant.

Temperature-dependence of K_{eq} (van't Hoff equation):

$$K_{eq}(T) = K_{eq}(T_1) \exp\left[\frac{\Delta H_{rxn}^\circ}{R} \left(\frac{1}{T_1} - \frac{1}{T}\right)\right] \quad (47)$$

Note that Equation 47 assumes that ΔH_{rxn}° does not depend on temperature.

For gas phase reactions, the concentration equations also depend on T . See Section 12.

12 Concentration Equations

Concentration is Moles $\div$ Volume, i.e.,

$$C = \frac{N}{V} \quad (48)$$

in a batch system and

$$C = \frac{F}{v} \quad (49)$$

in a flow system.

Concentration has units of moles per volume. $1 \text{ M} = 1 \text{ mol/liter} = 1 \text{ mol/dm}^3 = 1 \text{ kmol/m}^3 = 0.062 \text{ lbmol/ft}^3$.

12.1 Liquid Phase

Volume and volumetric flowrate are assumed constant.

$$C_A = \frac{N_A}{V_0} \quad (50)$$

in a batch system and

$$C_A = \frac{F_A}{v_0} \quad (51)$$

in a flow system.

In terms of X :

$$C_B = C_{LR,0} \left(\Theta_B - \frac{\nu_B}{\nu_{LR}} X \right) \quad (52)$$

12.2 Gas Phase

All equations in this section assume ideal gas.

The concentration of species A,

$$C_A = \frac{P_A}{RT} = \frac{y_A P}{RT} \quad (53)$$

where P_A is the partial pressure of A, y_A is the mole fraction of A, and P is the total pressure.

Change in volumetric flowrate for ideal gases (general equation):

$$v = v_0 \left(\frac{F_T}{F_{T0}} \right) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right) \quad (54)$$

Change in concentration for ideal gases (general equation):

$$C_B = C_{T0} \left(\frac{F_B}{F_T} \right) \left(\frac{P}{P_0} \right) \left(\frac{T_0}{T} \right) \quad (55)$$

Change in partial pressure for ideal gases (general equation):

$$P_B = P_{T0} \left(\frac{F_B}{F_{T0}} \right) \left(\frac{P}{P_0} \right) \quad (56)$$

Note that P_B does not comprise a temperature term. The only influence to total pressure is drag, which is tracked by the $\left(\frac{P}{P_0} \right)$ term. Partial pressure is additionally influenced by mole fraction, which is tracked by the $\left(\frac{F_B}{F_{T0}} \right)$ term.

Change in moles: For the reaction $aA + bB \longrightarrow cC + dD$, where A is the limiting reactant:

$$\delta = \frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1 \quad (57)$$

δ must be written in terms of the limiting reactant.

Mole change parameter:

$$\epsilon = y_{\text{LR},0} \delta \quad (58)$$

where $y_{\text{LR},0}$ is the inlet/initial mole fraction of the limiting reactant.

Change in volumetric flowrate for ideal gases (for cases that can be written in terms of conversion, X):

$$v = v_0 (1 + \epsilon X) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right) \quad (59)$$

Change in concentration for ideal gases (for cases that can be written in terms of conversion, X):

$$C_{\text{B}} = \frac{C_{\text{LR},0} \left(\Theta_{\text{B}} - \left(\frac{\nu_{\text{B}}}{\nu_{\text{LR}}} \right) X \right)}{1 + \epsilon X} \left(\frac{P}{P_0} \right) \left(\frac{T_0}{T} \right) \quad (60)$$

where ν_j are stoichiometric coefficients, which by definition are negative for reactants and positive for products.

12.3 Volume of an Ideal Gas and Concentration

1 mol of an ideal gas at 0°C and 1 atm occupies $22.4 \text{ dm}^3 \left(0.0446 \frac{\text{mol}}{\text{dm}^3} \right)$.

1 lbmol of an ideal gas at 32°F and 1 atm occupies $359 \text{ ft}^3 \left(0.00279 \frac{\text{lbmol}}{\text{ft}^3} \right)$.

13 Heat Exchange Scenarios

13.1 Adiabatic

Adiabatic energy balance equations for CSTRs, PFRs, batch reactors, and PBRs: These all have the same equations. This is because in the absence of heat transfer, the only contributions to the energy balance are the heat of reaction and the sensible heat changes of the chemicals entering and leaving the reactor. These properties are state functions, which means that they only depend on the inlet and outlet (or initial and final) conditions, not the path that was taken to get there.

Conversion in terms of temperature:

$$X_{\text{EB}} = \frac{\sum_i \Theta_i C_{Pi} (T - T_0)}{-\Delta H_{rxn}^\circ (T_{ref}) - \Delta C_P (T - T_{ref})} \quad (61)$$

where ΔH_{rxn} should be written with respect to the limiting reactant.

Temperature in terms of conversion:

$$T = \frac{(-\Delta H_{rxn}^\circ (T_{ref})) X_{\text{EB}} + \sum_i \Theta_i C_{Pi} T_0 + X_{\text{EB}} \Delta C_P T_{ref}}{\sum_i \Theta_i C_{Pi} - X_{\text{EB}} \Delta C_P} \quad (62)$$

where ΔH_{rxn} should be written with respect to the limiting reactant.

13.2 Constant T_a

When the cooling fluid temperature is constant, you can use the equations in Section 8.

13.3 T_a is Not Constant

13.3.1 CSTR

Use the heat transfer balance for a perfectly mixed vessel:

$$\dot{Q} = \dot{m}_c C_{Pc} (T_{a0} - T) \left[1 - \exp \left(-\frac{UA}{\dot{m}_c C_{Pc}} \right) \right] \quad (63)$$

13.3.2 Tubular Reactors (PFR, PBR)

For a co-current flow:

$$\frac{dT_a}{dV} = \frac{Ua(T - T_a)}{\dot{m}_c C_{Pc}} \quad (64)$$

where $T_a = T_{a0}$ at $V = 0$.

For counter-current flow:

$$\frac{dT_a}{dV} = \frac{Ua(T_a - T)}{\dot{m}_c C_{Pc}} \quad (65)$$

where $T_a = T_{a0}$ at $V = V_{final}$. The counter-current case must be solved iteratively until this condition is met.

13.4 Batch Reactor

A transient energy balance must be written on the cooling fluid and coupled with the batch reactor mole and energy balance equations.

14 Residence Time Distribution

14.1 Tracer Experiment Analysis

The residence time distribution is

$$E(t) = \frac{C(t)}{\int_0^\infty C(t)dt} \quad (66)$$

where $C(t)$ is concentration vs. time data taken during a tracer experiment. $E(t)$ indicates the fraction of fluid that exhibits a particular residence time. $C(t)$ is typically integrated numerically, although regression and interpolation are sometimes used.

The cumulative residence time distribution is

$$F(t) = \int_0^t E(t')dt' \quad (67)$$

$0 \leq F(\tau) \leq 1$, and $F(\infty) = 1$ (i.e., the sum of all probabilities = 1).

The mean residence time t_m is

$$t_m = \int_0^\infty tE(t)dt \quad (68)$$

The residence time in an ideal PFR is

$$E(t) = \delta(t - \tau) \quad (69)$$

which is the Dirac delta function. In other words, all fluid in the PFR has a single residence time τ .

The residence time in an ideal CSTR is

$$E(t) = \frac{\exp\left(-\frac{t}{\tau}\right)}{\tau} \quad (70)$$

The ideal CSTR residence time distribution reflects the phenomenon that some fluid elements leave the reactor almost as soon as they enter, while others remain in the reactor indefinitely. The ideal CSTR is a model; most real reactors exhibit different residence time distributions.

The dimensionless residence time distribution in an ideal CSTR is

$$E(\Theta) = \tau E(t) = \exp(-\Theta) \quad (71)$$

where $\Theta = \frac{t}{\tau}$. The advantage to using $E(\Theta)$ is that all ideal CSTR residence time distributions collapse onto a single curve.

14.2 Models for Estimating Reactor Conversion

For an ideal CSTR or PFR, $t_m = \tau$.

Non-ideal reactors can be modeled in a variety of ways. In this class, we use two extremes of mixing:

The *Segregation Model* assumes that fluid elements only “see” other fluid elements with the same residence time and do not mix with fluid elements with different residence times. In this model:

$$X_{seg} = \int_0^\infty X(t)E(t)dt \quad (72)$$

where $X(t)$ is obtained using the constant pressure batch reactor mole balance in coordination with the reaction rate law.

The *Maximum Mixedness* model assumes that the fluid is mixed at the molecular level, meaning that all fluid elements “see” all other fluid elements, regardless residence time. This model is similar to perfect mixing, with the difference being the residence time distribution. In the maximum mixedness model:

$$\frac{dX}{dz} = \frac{-r_A}{C_{A0}} - \frac{E(\bar{T} - z)}{1 - F(\bar{T} - z)} X \quad (73)$$

where z is the time variable, $\bar{T}$ is the maximum time for which $C(t)$ data exists, and A is the limiting reactant. Equation 73 hence integrates backwards from $z = \bar{T}$ to $z = 0$ with the boundary condition that $X(\bar{T}) = 0$. Equation 73 is typically solved using a combination of interpolation or regression in coordination with numerical integration.

15 Numerical Methods Cheat Sheet

- **Interpolation:** Estimate values between data points.
- **Regression:** Fit a model to data.
- **Numerical Integration:** Evaluate integrals.
- **Root Finding:** Solve $f(x) = 0$.
- **ODE Solver:** Solve differential equations such as $\frac{dX}{dt}$, $\frac{dF_A}{dV}$, and $\frac{dT}{dV}$.

16 Other Useful Equations

General mole balance:

$$F_{j0} - F_j + \int^V r_j dV = \frac{dN_j}{dt} \quad (74)$$

Mole balance that accounts for diffusion in one dimension:

$$D_{AB} \frac{d^2 C_A}{dz^2} - U_z \frac{dC_A}{dz} + r_A = 0 \quad (75)$$

Ergun equation (full and general equation):

$$\frac{dP}{dz} = -\frac{G}{\rho g_c D_P} \left(\frac{1-\phi}{\phi^3} \right) \left[\frac{150(1-\phi)\mu}{D_P} + 1.75G \right] \quad (76)$$

Simplified Ergun equation (collapses some parameters into a single parameter, general equation):

$$\frac{dp}{dW} = -\frac{\alpha}{2p} \frac{T}{T_0} \frac{F_T}{F_{T0}} \quad (77)$$

Simplified Ergun equation (collapses some parameters into a single parameter, for cases that can be written in terms of a single independent variable, X):

$$\frac{dp}{dW} = -\frac{\alpha}{2p} (1 + \epsilon X) \frac{T}{T_0} \quad (78)$$

Pseudo-Steady-State Hypothesis (PSSH):

$$r_{A^*} = \sum_{i=1}^n r_{iA^*} = 0 \quad (79)$$

where A^* is an active intermediate.

General energy balance:

$$\frac{d\hat{E}_{\text{sys}}}{dt} = \dot{Q} - \dot{W}_s + \sum_{i=1}^n F_{i0} H_{i0} - \sum_{i=1}^n F_i H_i \quad (80)$$

Note that this equation neglects kinetic and potential energy, since they are typically significantly smaller than $\dot{Q}$ and $\sum_{i=1}^n F_{i0} H_{i0} - \sum_{i=1}^n F_i H_i$.

Thiele modulus for diffusion limited catalytic systems

ϕ_n = square root ratio of reaction and diffusion rates

where n is the reaction order.

$$\phi_n = R \sqrt{\frac{k_n C_{As}^{n-1}}{D_e}} \quad (81)$$

where s indicates conditions at the pellet surface.

Internal effectiveness factor for diffusion limited catalytic systems

$$\eta = \frac{\text{actual rate of overall reaction}}{\text{rate of reaction that would result if } C_A \text{ inside the pellet} = C_{As}} \quad (82)$$

For large values of ϕ_n :

$$\eta = \sqrt{\frac{2}{n+1}} \frac{3}{\phi_n} \quad (83)$$

17 Unit Conversion Factors

17.1 Volume

$$\begin{array}{ll} 1 \text{ cm}^3 = 0.001 \text{ dm}^3 & 1 \text{ in}^3 = 0.0164 \text{ dm}^3 \\ 1 \text{ fluid oz} = 0.0296 \text{ dm}^3 & 1 \text{ ft}^3 = 28.32 \text{ dm}^3 \\ 1 \text{ m}^3 = 1000 \text{ dm}^3 & 1 \text{ U.S. gallon} = 3.785 \text{ dm}^3 \\ 1 \text{ L} = 1 \text{ dm}^3 & 1 \text{ ft}^3 = 7.482 \text{ gal} \end{array}$$

17.2 Length

$$1 \text{ \AA} = 10^{-8} \text{ cm} \quad 1 \text{ dm} = 10 \text{ cm}$$

$$1 \text{ }\mu\text{m} = 10^{-4} \text{ cm} \quad 1 \text{ in} = 2.54 \text{ cm}$$

$$1 \text{ ft} = 30.48 \text{ cm} \quad 1 \text{ m} = 100 \text{ cm}$$

17.3 Pressure

$$1 \text{ torr} = 0.13333 \text{ kPa} \quad 1 \text{ torr} = 1 \text{ mmHg}$$

$$1 \text{ in. H}_2\text{O} = 0.24886 \text{ kPa} \quad 1 \text{ Pa} = 1 \frac{\text{N}}{\text{m}^2}$$

$$1 \text{ in. Hg} = 3.3843 \text{ kPa} \quad 1 \text{ atm} = 101.33 \text{ kPa}$$

$$1 \text{ psia} = 6.8943 \text{ kPa} \quad 1 \frac{\text{megadyne}}{\text{cm}^2} = 100 \text{ kPa}$$

17.4 Temperature

$$^{\circ}\text{F} = 1.8 \times ^{\circ}\text{C} + 32 \quad ^{\circ}\text{R} = ^{\circ}\text{F} + 459.69$$

$$\text{K} = ^{\circ}\text{C} + 273.15 \quad ^{\circ}\text{R} = 1.8 \times \text{K}$$

17.5 Mass

$$1 \text{ lb}_\text{m} = 454 \text{ g} \quad 1 \text{ grain} = 0.0648 \text{ g}$$

$$1 \text{ oz} = 28.35 \text{ g} \quad 1 \text{ U.S. short ton} = 908,000 \text{ g}$$

17.6 Energy

$$1 \text{ Btu} = 1055.06 \text{ J} \quad 1 \text{ cal} = 4.1868 \text{ J}$$

$$1 \text{ L} \cdot \text{atm} = 101.34 \text{ J} \quad 1 \text{ hp} \cdot \text{h} = 2.6806 \times 10^6 \text{ J}$$

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J} \quad 1 \text{ kg} \cdot \text{m}^2/\text{s}^2 = 1 \text{ J}$$

17.7 Viscosity

$$1 \text{ poise} = 1 \frac{\text{g}}{\text{cm} \cdot \text{s}} \quad 1 \text{ centipoise} \text{ (1 cp)} = 1 \text{ mPa} \cdot \text{s} = 10^{-3} \text{Pa} \cdot \text{s}$$

17.8 Force

$$1 \text{ dyne} = 1 \frac{\text{g} \cdot \text{cm}}{\text{s}^2} \quad 1 \text{ Newton} = 1 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$
$$1 \text{ lb}_f = 4.448 \text{ N} \quad 1 \text{ lb}_f = 32.174 \frac{\text{ft} \cdot \text{lb}_m}{\text{s}^2}$$

17.9 Power

$$1 \text{ Watt (1 W)} = 1 \frac{\text{J}}{\text{s}} \quad 1 \text{ hp} = 746 \text{ W}$$

17.10 Work

Work = Force $\times$ Distance

Work = $\int P dV$. For constant P, $W = P \times V$.

17.11 Gravity and the Gravitational Conversion Factor (g_c)

Gravitational constant: $g = 32.174 \frac{\text{ft}}{\text{s}^2}$

Gravitational conversion factor: $g_c = \frac{32.174 \text{ ft} \cdot \text{lb}_m}{\text{s}^2 \cdot \text{lb}_f}$

g_c in SI units is equal to 1.

18 Useful Definite Integrals

$$\int_0^x \frac{dX}{1-X} = \ln \left(\frac{1}{1-x} \right) \quad (84)$$

$$\int_{x_1}^{x_2} \frac{dX}{(1-X)^2} = \frac{1}{1-x_2} - \frac{1}{1-x_1} \quad (85)$$

$$\int_0^x \frac{dX}{(1-X)^2} = \frac{x}{1-x} \quad (86)$$

$$\int_0^x \frac{dX}{1+\epsilon X} = \frac{1}{\epsilon} \ln(1+\epsilon x) \quad (87)$$

$$\int_0^x \frac{(1+\epsilon X) dX}{1-X} = (1+\epsilon) \ln \left(\frac{1}{1-x} \right) - \epsilon x \quad (88)$$

$$\int_0^x \frac{(1+\epsilon X) dX}{(1-X)^2} = \frac{(1+\epsilon) x}{1-x} - \epsilon \ln \left(\frac{1}{1-x} \right) \quad (89)$$

$$\int_0^x \frac{(1+\epsilon X)^2 dX}{(1-X)^2} = 2\epsilon(1+\epsilon) \ln(1-x) + \epsilon^2 x + \frac{(1+\epsilon)^2 x}{1-x} \quad (90)$$

$$\int_0^x \frac{dX}{(1-X)(\Theta_B - x)} = \frac{1}{\Theta_B - 1} \ln \left(\frac{\Theta_B - x}{\Theta_B(1-x)} \right) \quad (91)$$

where $\Theta_B \neq 1$.

$$\int_0^w (1-\alpha W)^{\frac{1}{2}} dW = \frac{2}{3\alpha} \left[1 - (1-\alpha w)^{\frac{3}{2}} \right] \quad (92)$$

$$\int_0^x \frac{dX}{aX^2 + bX + c} = \frac{-2}{2ax + b} + \frac{2}{b} \quad (93)$$

where $b^2 = 4ac$.

$$\int_0^x \frac{dX}{aX^2 + bX + c} = \frac{1}{a(p - q)} \ln \left(\frac{q x - p}{p x - q} \right) \quad (94)$$

for $b^2 > 4ac$, where p and q are the roots of the equation $aX^2 + bX + c$, i.e., $p, q = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

$$\int_0^x \frac{a + bX}{c + gX} dX = \frac{bx}{g} + \frac{ag - bc}{g^2} \ln \left(\frac{c + gx}{c} \right) \quad (95)$$

19 Symbol Definitions

Symbol	Example units
A = any reactive species	-
A = area	m^2
A = pre-exponential (frequency) factor	same as k
A = Helmholtz free energy	kJ/mol
a = tubular reactor heat exchange area	m^2/m^3
B = any reactive species	-
C = concentration	mol/L
C_P = heat capacity	$\frac{\text{J}}{\text{mol}\cdot\text{K}}$
C = any reactive species	-
D_P = diameter of catalyst particle	m
D_e = effective diffusivity	m^2/s
D = any reactive species	-
D = desired product	-
E = activation energy	kJ/mol
ΔE^{act} = activation energy	kJ/mol
E = system energy	kJ/mol
$E(t)$ = residence time distribution function	min^{-1}
F = molar flowrate	mol/s

F_T = total molar flowrate	mol/s
$F(t)$ = cumulative residence time distribution	unitless
F = Helmholtz free energy	kJ/mol
g_c = gravitational conversion factor	$\frac{\text{lb}_m \cdot \text{ft}}{\text{s}^2 \cdot \text{lb}_f}$
G = Gibbs free energy	kJ/mol
G = superficial mass velocity = ρu	$\frac{\text{kg}}{\text{m}^2 \cdot \text{s}}$
$G(T)$ = heat generated	kJ/mol
H = enthalpy	kJ/mol
ΔH_{rxn} = heat of reaction	kJ/mol
K_p = pressure equilibrium constant	atm $^\delta$
K_c = concentration equilibrium constant	$\frac{\text{mol}^\delta}{\text{L}^\delta}$
k = rate constant	$\frac{\text{L}^n}{\text{mol}^n \cdot \text{s}}, n = \text{reaction order} - 1$
LR = limiting reactant	-
$\dot{m}_c$ = mass flowrate of coolant	kg/s
N = number of moles	mol
P = pressure	atm
p = reduced pressure = $\frac{P}{P_0}$	unitless
p = momentum	N · s
Q = heat	kJ
Q = reaction quotient	unitless
Q_g = heat generated	kJ/mol

Q_r = heat removed	kJ/mol
R = gas constant	$\frac{\text{J}}{\text{mol}\cdot\text{K}}$ or $\frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}}$
R = rate of transport through a membrane	$\frac{\text{mol}}{\text{L}\cdot\text{s}}$
R = radius of a catalyst pellet	m
$R(T)$ = heat removed	kJ/mol
r = rate	$\frac{\text{mol}}{\text{dm}^3\cdot\text{s}}$
r' = rate	$\frac{\text{mol}}{\text{kg catalyst}\cdot\text{s}}$
S = entropy	$\frac{\text{J}}{\text{mol}\cdot\text{K}}$
S = selectivity	moles of D/moles of U
T = temperature	K
T_a = ambient or cooling fluid temperature	K
T_R = reaction temperature	K
T_{ref} = reference temperature, usually 298 K	K
$\bar{T}$ = maximum time	min
t = time	s
t_m = mean residence time	min
U = overall heat transfer coefficient	$\frac{\text{J}}{\text{m}^2\cdot\text{s}\cdot\text{K}}$
Ua = volumetric heat transfer coefficient ($= U \frac{dA}{dV}$)	$\frac{\text{J}}{\text{m}^3\cdot\text{s}\cdot\text{K}}$
U = velocity	m/s
U = internal energy	kJ/mol
U = undesired product	-

u = velocity	m/s
V = volume	dm ³
v = volumetric flowrate	dm ³ /min
v = velocity	m/s
W = weight of catalyst	kg
W_s = shaft work	kJ/mol
X = conversion	unitless
X_e = equilibrium conversion	unitless
x = mole fraction (liquid phase)	moles of A per mole
Y = yield	moles of D/moles of A
y = mole fraction (vapor phase)	moles of A per mole
z = reactor length	m
z = time	min
α = pressure drop parameter	kg ⁻¹
β = approach to equilibrium	unitless
δ = Mole change parameter	moles per mole of A
δ = Dirac delta	-
ϵ = Mole change parameter	unitless
Θ = ratio of moles or molar flowrates	moles of B per mole of A
Θ = dimensionless time	unitless
μ = viscosity	$\frac{\text{kg}}{\text{m}\cdot\text{s}}$
ν = stoichiometric coefficient	unitless

ρ = density	kg/m ³
ρ_b = bulk density of catalyst	$\frac{\text{kg catalyst}}{\text{L of reactor bed}}$
τ = residence time	s
ϕ = void fraction of packed bed	unitless
ϕ_n = Thiele modulus for an n order reaction	unitless
ξ = extent of reaction	mol